Daneesh Naheem P

AI Engineer

+91-9061253681 | daneeshnaheem@gmail.com | linkedin.com/in/daneesh-naheem-606a9921a

Professional Summary

AI Engineer with a B.E. in Artificial Intelligence and Machine Learning from Yenepoya Institute of Technology (VTU), graduating with First Class with Distinction. Experienced in developing and deploying enterprise AI solutions, with expertise in Computer Vision, Machine Learning, Generative AI, Retrieval-Augmented Generation (RAG), and backend development. Proficient in Python, Java, Spring Boot, FastAPI, PyTorch, YOLO, OpenCV, Azure AI services, PostgreSQL, and React. Skilled in building scalable AI pipelines, intelligent document processing systems, real-time video analytics, and production-ready REST APIs, with a strong focus on model performance, system reliability, and practical problem-solving

Professional Experience

AI Engineer | Lenok Solutions, Kozhikode, Kerala *Dec 2025 – Present*

- Develop enterprise AI applications using Python, FastAPI, PostgreSQL, Docker, React, and PyTorch.
- Build multi-camera computer vision pipelines for real-time object detection, aircraft/GSE segmentation, tracking, and event analysis.
- Develop RAG applications using LangChain, Spring AI, Ollama, FAISS, and vector databases for enterprise document retrieval.
- Design REST APIs and telemetry services; optimize inference, tracking accuracy, and real-time processing reliability.

Projects

- **Lenok Airport CRM – AI Contract Extraction and Review**
  Python, FastAPI, Azure Document Intelligence, Microsoft Foundry, JavaScript, Docker
  - Built an evaluation-ready CRM contract review application for scanned PDFs, native PDFs, and DOCX files using Azure Document Intelligence and Microsoft Foundry.
  - Added structured field extraction, page citations, evidence checks, and a review screen for users to correct results before creating a CRM record.
- **Lenok TMS – Turnaround Management System**
  Python, YOLOv8-Seg, ByteTrack, PyTorch, FastAPI, OpenCV, React, PostgreSQL
  - Built a multi-camera AI pipeline to analyze aircraft and ground-support-equipment interactions in real time.
  - Improved aircraft segmentation using custom mask processing and OpenCV morphology; developed tracking and turnaround event logic with ByteTrack and finite-state machines.
  - Integrated PostgreSQL telemetry with a React dashboard for operational monitoring.
- **Enterprise Vision Extraction Engine – Airport Utility OCR**
  Java 21, Spring Boot, ONNX Runtime (RapidOCR), PostgreSQL, Spring Data JPA, REST APIs
  - Built a Spring Boot service to extract meter readings (kWh, dates, and times) from scratched and high-glare seven-segment LCD images.
  - Applied morphological dilation to reconnect broken LCD segments before ONNX OCR inference; used deterministic regular expressions to parse readings.
  - Added image-quality checks using Laplacian variance and pixel thresholds, plus API-key filtering and offline mobile synchronization support.
  - Stored searchable verification metadata in PostgreSQL and archived raw images on disk.

- **SmartLog Occupancy**
  Python, YOLO, YOLO26-Pose, OCSort, FastAPI, Docker, PostgreSQL
  - Developed real-time entry/exit counting using YOLO, YOLO26-Pose ankle-keypoint detection, OCSort, polygon zones, and RTSP camera streams.
  - Improved stream handling with multithreaded queues, frame dropping, FP16 inference, and dynamic resolution scaling; exposed live analytics through FastAPI, WebSockets, and PostgreSQL.
- **Airport Management RAG Chatbot**
  Java, Spring Boot, Spring AI, Ollama, PgVector
  - Built a local RAG chatbot for searching airport operations and compliance documents, using semantic embeddings, vector storage, LLM inference, and chat memory.

Education

- **B.E. in Artificial Intelligence and Machine Learning** – Yenepoya Institute of Technology *2021 – 2025*
  Visvesvaraya Technological University (VTU) – First Class with Distinction
- **Higher Secondary (DHSE)** – 87% *2021*
- **Secondary (Kerala Board)** – 77% *2019*

Internships & Training

Kodnest Technologies | Full-Stack Development Program *2025*

Completed practical training in Java, Python, SQL, HTML, CSS, JavaScript, and React.js. Covered OOP, DSA, Collections, REST APIs, JDBC, MySQL, Servlets, JSP, Hibernate, Spring Boot, and Maven.

Zidio Development | Data Science & Analytics Intern *Mar – May 2025*

Worked on machine-learning workflows, text/audio/facial emotion analysis, mood-tracking dashboards, and HR alert functionality for an AI-powered task optimization project.

NSDC | Data Engineering Intern *Oct – Dec 2024*

Practiced SQL queries and joins, supported data migration and ETL workflows using Talend Open Studio, and created Power BI dashboards for data reporting.

TORC Infotech | AI / Web Development Intern *Oct – Nov 2023*

Contributed Python-based recommendation and sentiment-analysis features and developed responsive web dashboard components.

Genesis Private Ltd. | Android Development Intern *Oct – Nov 2022*

Developed basic Android applications using Java and XML, including app-level functionality and local data handling.

Technical Skills

Programming: Python, Java, C, C++, SQL, HTML, CSS, JavaScript, TypeScript

AI / ML: Machine Learning, Deep Learning, CNN, Computer Vision, NLP, Generative AI, LLMs, RAG

Computer Vision: YOLOv8, YOLOv8-Seg, ByteTrack, OCSort, OpenCV, OCR, Object Detection, Segmentation, Tracking

Frameworks: PyTorch, TensorFlow, FastAPI, Flask, Spring Boot, Spring AI, LangChain, ONNX Runtime

Backend / APIs: REST APIs, WebSockets, Spring Data JPA, Hibernate, API-key filters

Databases: PostgreSQL, MySQL, MongoDB, SQLite, PgVector, FAISS

Tools: Docker, Git, GitHub, Ollama, Maven, Postman, React, Power BI, Pandas, NumPy